1. INTRODUCTION

1.1 OBJECTIVE

a) To structure KSRTC Mangalore bus timetable data in Microsoft Excel, taking origin, destination, service type, via place, and departure times.
b) To classify departure times into predefined time slots and integrate district-level population and urbanisation indicators to estimate relative crowd levels.
c) To develop a framework to assign overcrowding risk levels to different routes and time slots based on structured secondary data.
d) To create a Tableau-based dashboard to visualise estimated crowd levels and identify high and low demand time slots for better bus planning.
e) To ensure the analysis is practical, transparent, and reproducible using publicly available data without relying on real-time passenger tracking or sensitive operational datasets.

1.2 PROPOSED METHODOLOGY

The research considers bus routes listed in the official timetable for all types of services. Since the study uses **secondary data only**, every available route is treated as part of the sample.

Primary Data: Not applicable (no field observations or real-time measurements).
Secondary Data:
a) KSRTC Mangalore official timetables (bus stand notices, PDF schedules, or website)
b) Government census and urbanisation data for population indicators.
Microsoft Excel: For structuring data, categorisation, and initial calculations.
Python (Pandas, NumPy): For data cleaning, feature engineering, and classification of crowd risk levels.
Tableau: For creating an interactive dashboard that visualises route-wise crowd levels and recommends suitable travel time slots.

1.3 PROBLEM STATEMENT

People using buses depend on schedules to plan their journey. Though these schedules give information about the route and timings, they do not give the safest time periods or the crowd levels. People struggle with peak-hour congestion and crowded travel times.

Also, detailed passenger count data, real-time systems etc. are not publicly available, creating a gap between the required information for travel planning and the data accessible for analysis.

There is a need for an approach where publicly available timetable data can be used to find busy time slots where additional buses may be required and to support better bus frequency planning based on demand patterns.

1.4 SOCIAL RELEVANCE AND UNIQUENESS OF THE WORK

Public transport improves mobility, reduces traffic congestion, and ensures access to services. Overcrowding at peak time slots leads to discomfort, congestion, and reduced travel convenience. Crowd intensity pattern is important to improve public transport planning and service efficiency.

This research is socially relevant as it focuses on analyzing crowd intensity patterns in KSRTC bus services using data-driven approaches. The study helps identify busy time slots where passenger movement is high, which may support transport authorities to understand time slots where additional bus services may be required. The work also contributes toward improving service planning and passenger convenience through better understanding of demand patterns across routes and timings.

The uniqueness of this work lies in its timetable-driven and population-aware approach to analyze crowd intensity patterns. This study uses available timetable data along with district-level demographic indicators like population, urbanisation level, and interstate connectivity. The proposed framework is low-cost, transparent, interpretable, and easy to implement. It provides a practical analytical approach that can support transport authorities, researchers, and urban mobility studies without privacy or operational concerns.

2. REVIEW OF LITERATURE

2.1 RELATED WORK

[1]. Basha, SK. et al. (2025): This work involved training and validating models with timestamps, route numbers, and traffic conditions, and comparing performance to improve passenger boarding demand in bus transit by using SVM and DNNs, to forecast demand dynamically. AI-based models delivered high precision in predicting boarding demand.

[2]. Shrivastava, A. et al. (2024): This work applied transit segment relations (TSR) to capture ridership patterns and predicted crowding levels for different periods of the day. The model, validated with three months of ticketing data in Bhubaneswar, can guide passengers and transit agencies about expected crowding, helping them plan route choices.

[3]. V S, D. et al. (2022): This work involved IoT, hardware sensors, and deep learning models to estimate crowd density and provide travel information. Such systems can contribute to better use of travel time and help both passengers and travel agencies schedule appropriately.

[4]. Hsu, WY. et al. (2020): This work processed visual or sensor data to detect head regions and count passengers, evaluating performance across different scenarios. The model performed better than existing methods in estimating occupancy.

[5]. Marques, S. et al. (2022): This work compared models with and without spatial dependence, finding that models recognizing spatial differences provide better fit and improve predictions. It emphasized the importance of incorporating spatial context and demographic predictors into ridership estimates.

[6]. Pronello, C. et al. (2024): This work involved installing sniffing devices at stops, collecting probe request data, and training deep learning models to predict passenger counts with an average error of about 1.2 people. Low-cost sensing combined with ML yielded reliable counts for planning.

[7]. Talusan, JP. et al. (2022): This work used XGBoost and LSTM on millions of observations to perform trip- and stop-level prediction, showing that data fusion and advanced models outperform baseline statistics. Multi-source datasets improve accuracy.

[8]. Martí, P. et al. (2024): This study introduced a machine learning framework with multi-agent simulations to predict bus ridership and analyze transportation scenarios, using a comprehensive urban bus dataset to forecast passenger flows and evaluate service effectiveness.

[9]. Fontes, T. et al. (2020): This work applied a deep learning neural network (Multilayer Perceptron) to predict bus passenger demand based on transit ridership and weather conditions over a year, showing that incorporating meteorological variables reduced prediction errors significantly. It labelled ridership data with hour of day and day of week, improving demand forecasting accuracy.

[10]. Farahmand, Z. H. et al. (2023): This research used deep learning (Multilayer Perceptron networks) to forecast bus ridership accounting for weather, holidays, and special events using smart-card data in the Twente region of the Netherlands, finding that including weather features improves prediction accuracy on regular weekdays.

[11]. Yusuf, O. et al. (2025): This study developed a horizon-agnostic modelling framework using automated passenger count (APC) data and machine learning to forecast stop-level transit passenger counts, incorporating feature optimization and interpretable model outputs.

[12]. J Siswanto et al. (2024): This work proposed a deep learning LSTM model for predicting passenger numbers across four bus public transportation operators using a multi-region historical dataset from Auckland Transport, demonstrating how temporal LSTM models can capture passenger flow dynamics for future months.

[13]. Drisilda, S. et al. (2025): This paper presents a machine learning-based framework to predict hourly ridership at Bangalore Metropolitan Transport Corporation stops using historical bus stop data and SARIMAX models to capture daily patterns and autocorrelations.

[14]. NNS Amir et al. (2025): This research developed a ridership prediction system for rapid bus services in Kuantan and Penang using multi-feature approaches that integrate route and service variables to forecast passenger demand.

[15]. Mahsa Movaghar et al. (2025): This study proposed a machine learning framework to estimate ridership loss in the bus network of Stockholm during external crises (e.g., pandemic) by comparing regression and algorithmic models for historical ridership data.

[16]. Zeb, M. S. et al. (2024): This research applied LSTM-based models to forecast public transit ridership amidst COVID-19 by integrating seasonal, intervention, and pandemic statistical inputs, achieving high predictive accuracy.

[17]. Akgüngör, A. P., et al. (2025): This work compares multiple deep learning models, including LSTM and NARX, for modelling short-term passenger flows in bus and rail systems using weather and school data, finding that NARX produced the lowest forecast errors.

[18]. Torrepadula, F. R. di., et al. (2024): This article surveys machine learning methods for public transportation demand forecasting, summarizing varied ML solutions and data types used across studies.

[19]. Ke, L., et al. (2025): The work emphasizes the importance of having detailed ridership datasets for benchmarking and improving forecasting, scheduling and clustering of bus services. By providing route details, stop distances and transaction-level time granularity, the dataset supports high-resolution analysis of ridership patterns and to evaluate predictive models more precisely.

[20]. Yamamura, T. et al. (2023): This paper proposes a multiple-LSTM model that integrates past ridership, day of week, time section, weather, and precipitation to improve bus ridership prediction accuracy, demonstrating significant error reduction compared to existing methods.

3. METHODOLOGY

3.1 SYSTEM REQUIREMENTS

3.1.1 HARDWARE REQUIREMENTS

Hardware Component(s)	Specification
Processor	Intel Core i5 or equivalent
RAM	Minimum 8 GB
Storage	256 GB SSD / 500 GB HDD
System Type	64-bit Operating System
Display	14-inch or higher monitor
Input Devices	Keyboard and Mouse
Internet Connectivity	Required for dataset collection and Tableau Public publishing

3.1.2 SOFTWARE REQUIREMENTS

Software / Tool	Purpose
Windows 10 / 11	Operating System
Python 3.x	Data processing and analysis
Jupyter Notebook	Data preprocessing and feature engineering
Pandas Library	Dataset handling and manipulation
NumPy Library	Numerical operations
Tableau Public	Data visualization and dashboard creation
Microsoft Excel	Data cleaning and observation management
Visual Studio Code	Code editing and development
CSV Files	Dataset storage and integration

3.2 DATA COLLECTION AND PREPROCESSING

3.2.1 IMPLEMENTATION OF DATASET AND DATA SOURCES

The implementation of the dataset for this study was carried out using available KSRTC timetable information and district-level demographic data. The dataset was organized in Excel and CSV format containing information like source, destination, service class, via place, bus timings, and slot labels. This structured dataset was used as the primary source for further pre-processing and analysis.

Sl.No	From	Destination	Service Class	Via Place	Time	Slot Label
1	Mangaluru	Dharmasthala	Limited	BC Road, Belthangadi	05:20:00	2 AM - 6 AM
2	Mangaluru	Dharmasthala	Limited	BC Road, Belthangadi	05:50:00	2 AM - 6 AM
3	Mangaluru	Dharmasthala	Limited	BC Road, Belthangadi	06:00:00	6 AM - 10 AM
4	Mangaluru	Dharmasthala	Limited	BC Road, Belthangadi	06:20:00	6 AM - 10 AM
5	Mangaluru	Dharmasthala	Limited	BC Road, Belthangadi	06:45:00	6 AM - 10 AM

Fig 3.1 Structured KSRTC Timetable Dataset with Time Slot Information

The timetable information was collected and organized in CSV format for easy pre-processing and analysis. Microsoft Excel and Python libraries such as Pandas were used to handle the dataset and to perform data integration tasks.

Column Name	Description
Sl.No.	Serial number assigned to each record
From	Source location from where the bus starts
Destination	Final destination of the route
Service Class	Category of bus service
Via Place	Intermediate stops covered by the route
Time	Scheduled departure time
Slot Label	Categorized time interval based on bus timing

Table 3.1 Description of Original Dataset Columns

District-level demographic data was collected from available statistical and census-based sources. These were merged with the KSRTC timetable dataset using common district-level mappings.

District	State	Population_Millions	Urbanisation Level	Region_Type	Interstate_Flag
Bangalore	Karnataka	10.96	Very High	Top Urban	No
Pune	Maharashtra	10.48	Very High	Top Urban	Yes
Mumbai Suburban	Maharashtra	10.40	Very High	Top Urban	Yes
Belagavi	Karnataka	5.44	Very High	Top Urban	No
Hyderabad	Telangana	4.28	Very High	Top Urban	Yes

Fig 3.2 District-Level Demographic Dataset

To integrate demographic information with the timetable, destination locations were mapped to their corresponding districts to link each route with district-level demographic attributes. The destination-to-district mapping was performed manually and verified using available geographical references.

Destination	District
Adyar Padavu	Dakshina Kannada
Amingad	Bagalkot
Ankalimata, Talekat	Raichur
Arsikere	Hassan
B.C.Road	Dakshina Kannada

Fig 3.3 Destination to District Mapping Dataset

Column Name	Description
Destination	Destination from the timetable
District	District mapped to the destination

Table 3.2 Description of Destination Mapping Columns

After integration, the final merged dataset contained route-level, time-slot, and demographic information required for further analysis and visualization.

Attribute Name	Description
From	Source location from where the bus starts
Destination	Final destination of the route
Service Class	Category of bus service
Via Place	Intermediate stops covered by the route
Time	Scheduled departure time
Slot Label	Categorized time interval based on bus timing
District	District mapped to the destination
State	State corresponding to the mapped district
Population_Millions	Population value of the district in millions
Urbanisation_Level	Categorized urbanisation level of the district
Region_Type	Classification of district as top urban, mid urban, semi urban etc..
Interstate_Flag	Indicates whether the route is interstate or intrastate. Intrastate – No, Interstate - Yes

Table 3.3 Final Merged Dataset Attributes

The final dataset was stored in CSV format and used for preprocessing, analysis, and Tableau dashboard visualization.

From	Destinatic	Service Cl	Via Place	Time	Slot Label	District	State	Populatio	Urbanisati	Region_Ty	Interstate
Mangalur	Dharmastl	Limited	BC Road, E	05:20:00	2 AM - 6 A	Dakshina	Karnataka	2.38	High	Mid Urbar	No
Mangalur	Dharmastl	Limited	BC Road, E	05:50:00	2 AM - 6 A	Dakshina	Karnataka	2.38	High	Mid Urbar	No
Mangalur	Dharmastl	Limited	BC Road, E	06:00:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No
Mangalur	Dharmastl	Limited	BC Road, E	06:20:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No
Mangalur	Dharmastl	Limited	BC Road, E	06:45:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No

Fig 3.3 Final Merged CSV Dataset

3.3 FEATURE ENGINEERING AND CROWD RISK SCORING

Feature engineering was performed to transform the collected timetable and demographic data into analytical attributes suitable for crowd intensity analysis. Numerical values were assigned to categorical attributes such as urbanisation level, interstate connectivity, and time-slot intervals to support the scoring process.

The time-slots were created based on observed frequency patterns across different time periods of the day. Peak-hour slots with higher bus frequency were assigned high scores, while late-night and early-morning slots were assigned low scores.

Time Slot	No. of Buses per slot	Time Score	Frequency of Buses
2 AM - 6 AM	27	1	<50
6 AM - 10 AM	130	3	>100
10 AM - 2 PM	123	3	>100
2 PM - 6 PM	142	3	>100
6 PM - 10 PM	90	2	50-100
10 PM - 2 AM	29	1	<50

Table 3.4 Time Slot Scoring and Bus Frequency Classification

Urbanisation levels and regional classifications were also converted into numerical representations using district-level demographic observations. Interstate routes were additionally identified using binary flag values.

Population_Million (Range)	Urbanisation Level	Region Type	Urban_Level_Num
>2.9	Very High	Top Urban	5
2.2-2.8	High	Mid Urban	4
2.0-2.1	Moderate	Semi Urban	3
1.6-1.9	Medium	Semi Rural	2
1.3-1.5	Low	Mid Rural	1
0.0-1.2	Very Low	Top Rural	0

Table 3.5.1 Urbanisation Level

Interstate_Flag	Interstate_Flag_Num	Category
Yes	1	Interstate
No	0	Intrastate

Table 3.5.2 Interstate Flag Mapping

A weighted crowd risk scoring approach was then applied using demographic and timetable-based indicators. Population, urbanisation level, interstate connectivity, and time-slot scores were combined to calculate the final Crowd_Risk_Score for each record.

Crowd_Risk_Score = (Population * 0.3) + *(Urban_Level_Num * 0.2) + (Interstate_Flag * 0.2) + (Time Score * 0.3)

Equation 3.1 Crowd Risk Score Formula

Based on the calculated score values, routes were classified into Low, Medium, and High crowd intensity levels.

Crowd Risk Score Range	Crowd Risk Level
0.0-1.2	Very Low
1.3-1.5	Low
1.6-1.9	Medium
2.0-2.1	Moderate
2.2-2.8	High
>2.9	Very High

Table 3.6 Crowd Risk Level Classification

The engineered features and scoring mechanism helped identify peak crowd intensity patterns across different KSRTC routes and time slots.

3.4 PSEUDOCODE

The pseudocode below represents the overall workflow used for crowd intensity analysis and crowd risk classification in the proposed system.

```
START
Load KSRTC timetable dataset
Load District-level Demographic dataset
Load Destination-to-District mapping dataset
Merge timetable dataset with mapping dataset
Merge demographic dataset with merged timetable dataset
Assign slot labels based on bus timings
Assign numerical scores for: Time slots, Urbanisation levels, Interstate connectivity
Calculate Crowd_Risk_Score using weighted scoring formula
IF Crowd_Risk_Score is > 2.9:
        Assign 'Very High' Crowd Risk Level
ELSE IF Crowd_Risk_Score is between 2.2 and 2.8:
        Assign 'High' Crowd Risk Level
ELSE IF Crowd_Risk_Score is between 2.0 and 2.1:
        Assign 'Moderate' Crowd Risk Level
ELSE IF Crowd_Risk_Score is between 1.6 and 1.9:
        Assign 'Medium' Crowd Risk Level
ELSE IF Crowd_Risk_Score is between 1.3 and 1.5:
        Assign 'Low' Crowd Risk Level
ELSE
        Assign 'Very Low' Crowd Risk Level
Store final processed dataset
Generate Tableau visualizations and dashboard
END
```

Figure 3.5 Pseudocode

The pseudocode summarizes the sequence of dataset integration, feature engineering, crowd risk score calculation, and dashboard visualization performed in this research work.

4. RESULTS

4.1 DATASET OVERVIEW

The final dataset was created by integrating KSRTC timetable information with district-level demographic data. The data has route information, bus timings, and district demographics to visualise crowd intensity analysis. The data consists of 542 records and 19 attributes.

	From	Destinatic	Service Cl	Via Place	Time	Slot Label	District	State	Populatio	Urbanisat	Region_Ty	Interstate	Start	End	Urban_Lev	Interstate	Time_Sco	Crowd_Ri	Crowd_Ris
0	Mangalun	Dharmastl	Limited	BC Road, I	05:20:00	2 AM - 6 A	Dakshina	Karnataka	2.38	High	Mid Urbar	No	02:00:00	06:00:00	4	0	1	1.814	Medium
1	Mangalun	Dharmastl	Limited	BC Road, I	05:50:00	2 AM - 6 A	Dakshina	Karnataka	2.38	High	Mid Urbar	No	02:00:00	06:00:00	4	0	1	1.814	Medium
2	Mangalun	Dharmastl	Limited	BC Road, I	06:00:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No	06:00:00	10:00:00	4	0	3	2.414	High
3	Mangalun	Dharmastl	Limited	BC Road, I	06:20:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No	06:00:00	10:00:00	4	0	3	2.414	High
4	Mangalun	Dharmastl	Limited	BC Road, I	06:45:00	6 AM - 10	Dakshina	Karnataka	2.38	High	Mid Urbar	No	06:00:00	10:00:00	4	0	3	2.414	High

Figure 4.1 Final Merged Dataset

Attribute Name	Description
From	Source location from where the bus starts
Destination	Final destination of the route
Service Class	Category of bus service
Via Place	Intermediate stops covered by the route
Time	Scheduled departure time
Slot Label	Categorized time interval based on bus timing
District	District mapped to the destination
State	State corresponding to the mapped district
Population_Millions	Population value of the district in millions
Urbanisation_Level	Categorized urbanisation level of the district
Region_Type	Classification of district as top urban, mid urban, semi urban etc..
Interstate_Flag	Indicates whether the route is interstate or intrastate. Intrastate – No, Interstate - Yes
Start	Start time of the assigned slot
End	End time of the assigned slot
Urban_Level_Num	Numerical representation of urbanisation level of the district
Interstate_Flag_Num	Numerical encoding of interstate connectivity (0 or 1)
Time_Score	Score assigned based on observed bus frequency within a time slot
Crowd_Risk_Score	Final weighted score used for crowd estimation intensity
Crowd_Risk_Level	Classified crowd intensity category

Table 4.1 Dataset Attribute Summary

4.2 DATA PREPROCESSING RESULTS

4.2.1 TIME SLOT ANALYSIS

Bus timings were divided into six time slots. Observational analysis showed that certain periods of the day had high bus frequency.

Time Slot	Number of Buses	Frequency Observation
2 AM - 6 AM	27	Low Frequency
6 AM - 10 AM	130	High Frequency
10 AM - 2 PM	123	High Frequency
2 PM - 6 PM	142	High Frequency
6 PM - 10 PM	90	Medium Frequency
10 PM - 2 AM	29	Low Frequency

Table 4.2 Time Slot Frequency Analysis

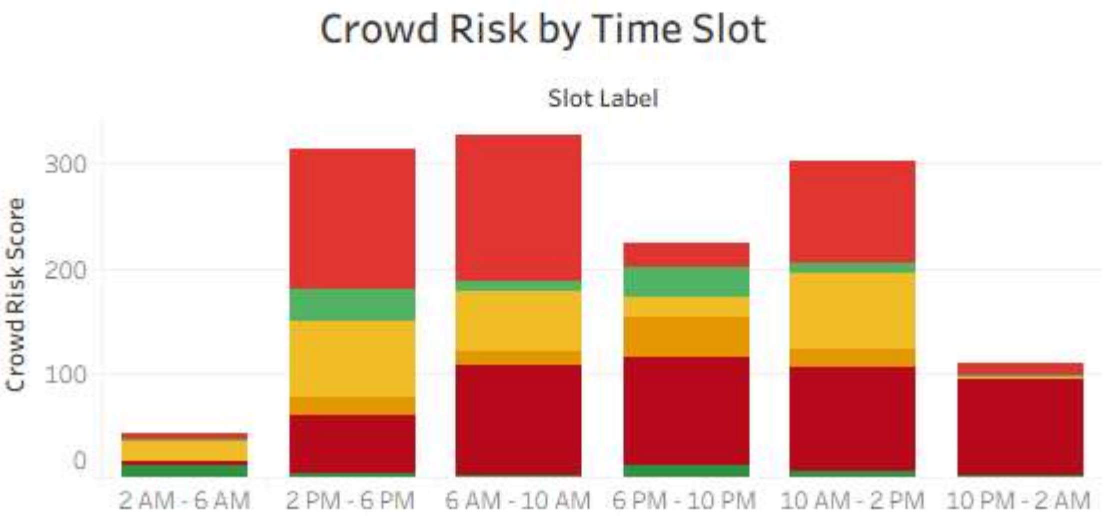

Chart 4.1 Bus Frequency by Time Slot

The results show that 6 AM–6 PM has the highest number of services, signifying high passenger movement during daytime.

4.2.2 FEATURE ENGINEERING RESULTS

Categorical variables were converted into numerical representations for crowd risk scoring. Urbanisation levels, interstate connectivity, and time-slot categories were transformed into analytical features.

Population_Million (Range)	Urbanisation Level	Region Type	Urban_Level_Num
>2.9	Very High	Top Urban	5
2.2-2.8	High	Mid Urban	4
2.0-2.1	Moderate	Semi Urban	3
1.6-1.9	Medium	Semi Rural	2
1.3-1.5	Low	Mid Rural	1
0.0-1.2	Very Low	Top Rural	0

Table 4.3 Urbanisation Level Mapping

Interstate_Flag	Interstate_Flag_Num	Category
Yes	1	Interstate
No	0	Intrastate

Table 4.4 Interstate Flag Mapping

Time Slot	Time Score
2 AM - 6 AM	1
6 AM - 10 AM	3
10 AM - 2 PM	3
2 PM - 6 PM	3
6 PM - 10 PM	2
10 PM - 2 AM	1

Table 4.5 Time Score Assignment

4.2.3 CROWD RISK CLASSIFICATION

A weighted crowd risk score was calculated using population, urbanisation level, interstate connectivity, and time-slot information.

Crowd_Risk_Score = (Population * 0.3) + *(Urban_Level_Num * 0.2) + (Interstate_Flag * 0.2) + (Time Score * 0.3)

Equation 4.1 Crowd Risk Score Formula

The resulting scores were classified into Very Low, Low, Medium, Moderate, High and Very High crowd risk categories.

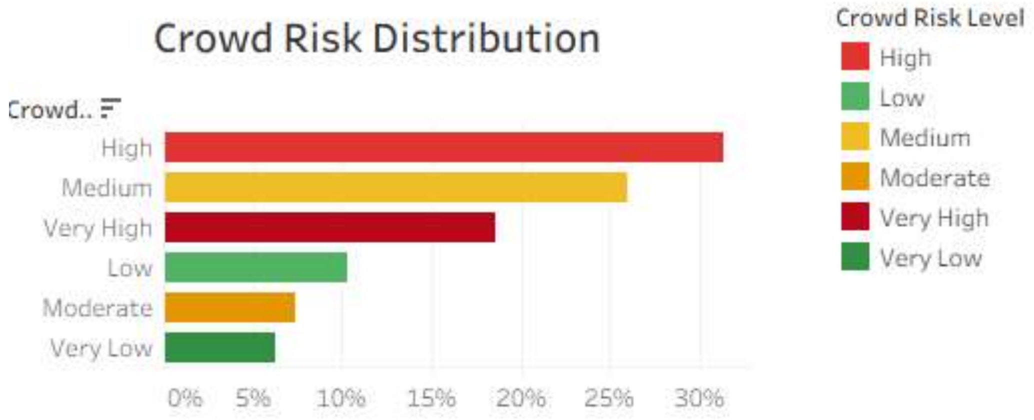

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Chart 4.2 Crowd Risk Distribution

Most routes fall under the High crowd risk category, while a smaller number of routes are classified as Very Low risk.

4.2.4 TABLEAU DASHBOARD VISUALISATIONS

An interactive dashboard was developed in Tableau to visualize crowd intensity patterns across routes, districts, and time slots.

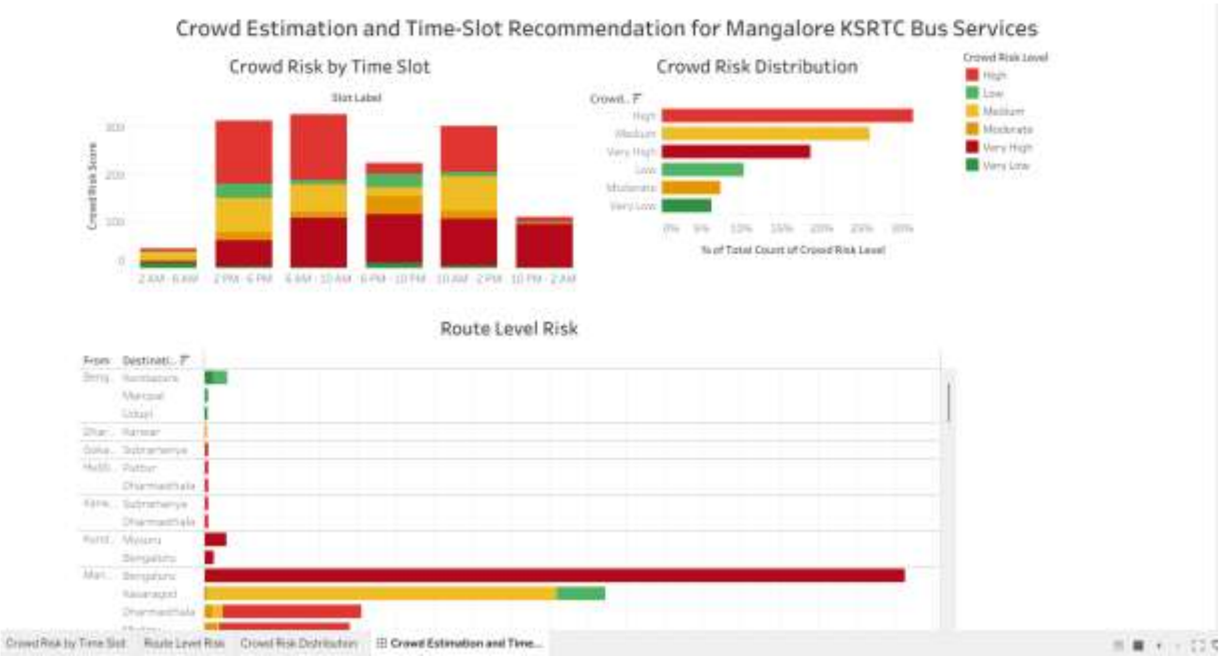

Figure 4.3 Tableau Dashboard

4.2.5 DISTRICT LEVEL ANALYSIS

District-level analysis was done to understand the distribution of crowd intensity.

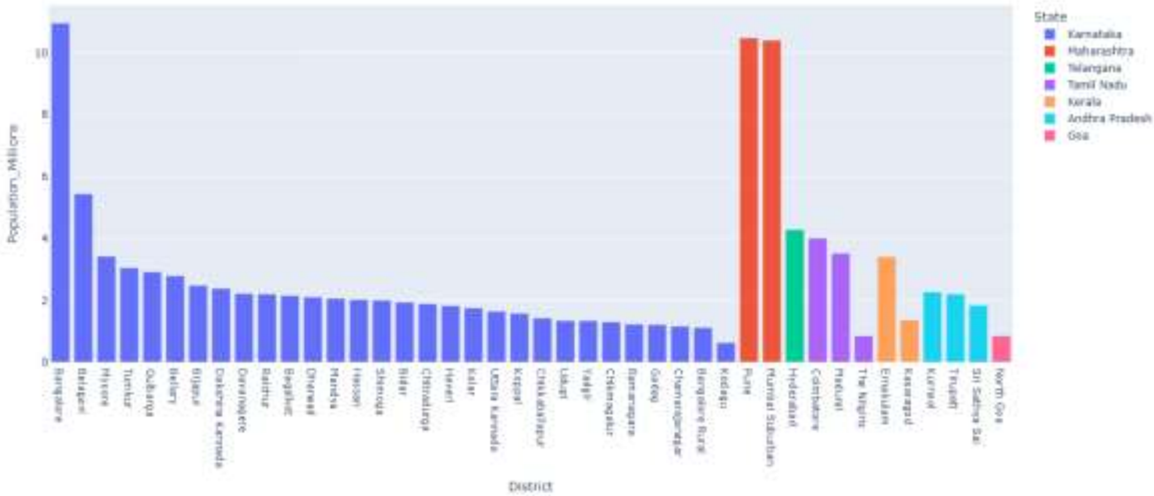

Chart 4.3 District Level Analysis

The analysis shows districts that contribute to high crowd intensity patterns due to demographic and route characteristics.

4.2.6 ROUTE LEVEL ANALYSIS

Route-level analysis was conducted to identify routes with high crowd risk scores.

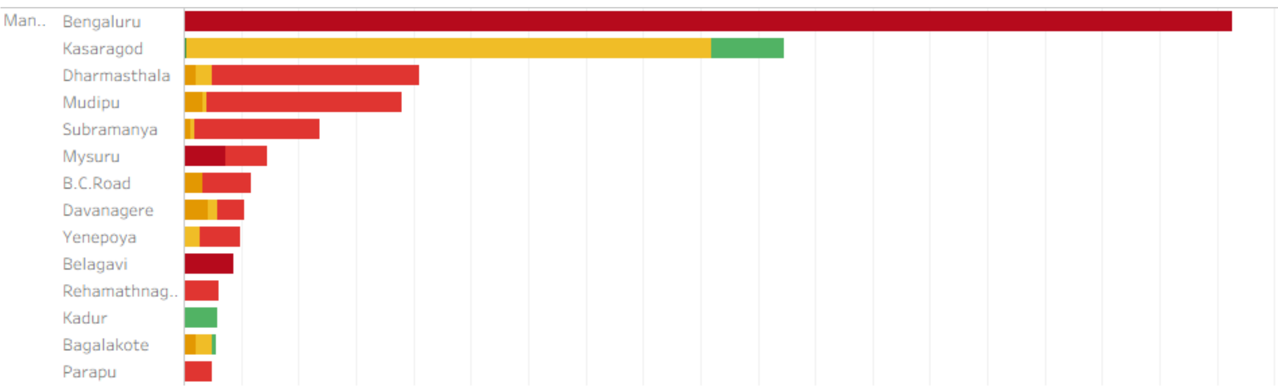

Chart 4.4 Route Level Risk Analysis

The visualization helps to identify routes that need monitoring or additional services when there is high-demand.

5. WEEKLY PROGRESS REPORTS

The Yenepoya Institute of Arts, Science, Commerce & Management
Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA
Register Number: 24MCA306
Date: 03|03|2026
Internal Guide's Name: MS SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Perform structured data exploration of "District_Info.csv" (exploratory data analysis) and analyse district population distribution.
→ Study/examine Urbanisation Level and Region Type patterns along with examining the impact of Interstate vs Non Interstate routes.
→ Generate basic visualisations like bar chart, histogram, pie chart, boxplot.
→ Prepare for calculating Crowd Risk Score based on population, urban level, interstate flag and time score.

Progress/Achievements for the current week:

→ District level dataset was restructured and cleared successfully.
→ Major districts were identified based on population in millions, urbanisation level, region type and interstate flag. Bangalore and Belagavi are among the top 5 districts that have more intra state buses operating and interstate also. Pune, Mumbai Suburban, Hyderabad and Coimbatore also perform well (operating more on bookings).
→ Other top districts are Mysore, Tumkur and Gulbarga (intrastate) and Madurai, Ernakulam (interstate)
→ In Karnataka, more buses are required in Bangalore, Belagavi, Mysore, Tumkur and Gulbarga districts to operate on city, interstate and village routes. Interstate routes can be managed provided additional buses need to be provided on festivals/ weekends.
→ Generated visualisations to understand data patterns.

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Yenepoya (Deemed to be University)

Future Work Plans (for the upcoming week):

Merge datasets into a single master dataset.
Remove duplicates and validate merged datasets for missing values, duplicates and data consistency.
Prepare a merged CSV file (single master dataset) for Crowd Risk Calculation & analysis.

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

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____

____

03|03|2026

Signature of the student

(with date)

Signature of the Internal Guide

(with date)

The Yenepoya Institute of Arts, Science, Commerce & Management
Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA Register Number: 24MCA306
Date: 12|03|2026
Internal Guide's Name: MS SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Merge datasets into a single master dataset.
→ Remove duplicates/Inconsistent records and validate the merged datasets for missing values, duplicates and data consistency.
→ Perform EDA (Exploratory Data Analysis) on merged dataset.
→ Prepare a merged CSV file for Crowd Risk Calculation & Analysis and time slot recommendations.

Progress/Achievements for the current week:

→ Successfully merged CSV files into "KSRTC_Merged.csv".
→ Duplicates were removed & verified unique mapping for all routes.
→ Merged dataset was validated: no missing values, duplicates; all entries consistent.
→ Preliminary visualisations were conducted to understand district population distribution, urbanisation levels, interstate vs intrastate route proportions. Identified high demand districts|routes.
→ Prepared merged dataset for Crowd Risk Score calculation & route-level time slot recommendation.

Future Work Plans (for the upcoming week):

→ Calculate Crowd Risk Score for all routes & integrate weekday/weekend patterns.
→ Perform analysis to identify high/low risk time slots.
→ Generate immediate CSV outputs for visualisations/dashboards.
→ Begin preparation of route-level recommendations based on time slot crowd patterns.

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

____

____

____

Leal Dsouza
12/03/2026
Signature of the student
(with date)

Signature of the Internal Guide
(with date)

The Yenepoya Institute of Arts, Science, Commerce & Management
Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA Register Number: 24MCA306
Date: 08|04|2026
Internal Guide's Name: MS SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Calculate Crowd Risk Score for all routes & integrate weekday/weekend patterns.
→ Perform analysis to identify high/low risk time slots.
→ Generate immediate CSV outputs for visualisations/dashboards
→ Begin preparation of route-level recommendations based on time slot patterns of crowding.

Progress/Achievements for the current week:

→ Successfully implemented Crowd Risk Score formula:–
Crowd Risk Score = (Population_Millions * 0.3) + (Urbanisation_Level * 0.2) + (Interstate_Flag * 0.2) + (Time_Score * 0.3) → [30% + 20% + 30% + 20% = 100%]
→ Generated crowd risk scores for all routes and time slots in the dataset.
→ Maintained model interpretability and kept weight distribution constant (60% demand factors and 40% contextual factors)
→ Ensure that dataset is ready for visualisation and recommendation.
→ Confirmed that peak time slots show relatively higher crowd risk values.

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Future Work Plans (for the upcoming week):

→ Perform visualization of Grand Risk Score distribution across routes, districts, time slots- using Tableau software.
→ Prepare structured outputs for creating the dashboard.
→ Analyse route-level patterns to determine safe travel periods.
→ Begin preparing for final outputs, research paper etc ———

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

____

____

____

Lea Dsouza
08/04/2026
Signature of the student

(with date)

Signature of the Internal Guide

(with date)

The Yenepoya Institute of Arts, Science, Commerce & Management
Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA   Register Number: 24MCA306
Date: 22|04|2026
Internal Guide's Name: MS SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Create categorization table for Population in Millions by considering population range, remarks, type and ID.
→ Population in millions is considered for districts only, not cities.
→ Also, Categorise the routes into interstate & intrastate through binary encoding.

Progress/Achievements for the current week:

→ Created a population(millions) categorisation table as follows:-

Population range	Remarks	Type	ID
>2.9	Very High	Top Urban	5
2.2-2.8	High	Mid Urban	4
2.0-2.1	Moderate	Semi Urban	3
1.6-1.9	Medium	Semi Rural	2
1.3-1.5	Low	Mid Rural	1
0.0-1.2	Very Low	Top Rural	0

→ Binary encoding

yes	1	Interstate
NO	0	intrastate

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Future Work Plans (for the upcoming week):

→ Create categorisation table for time scores by considering time slots, no. of buses per time slot, time score for each slot etc.

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

22/04/2026
Signature of the student

(with date)

Signature of the Internal Guide

(with date)

The Yenepoya Institute of Arts, Science, Commerce & Management
Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA
Date: 2/5/2026
Register Number: 24MCA306

Internal Guide's Name: Ms. SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Create classification/Categorisation tables for time scores ~~based on the~~ ~~as~~ by considering time slots, no. of buses per time slot, time score for each time slot etc ___

Progress/Achievements for the current week:

→ Created a time score table which is as follows:-

Time Score

Time Slot	No. of buses per slot	Time Score	Range of buses
2AM – 6AM	27	1	<50
6AM – 10AM	130	3	>100
10AM – 2PM	123	3	>100
2PM – 6PM	142	3	>100
6PM – 10PM	90	2	50–100
10PM – 2AM	29	1	<50

→ Here, it is found that the crowded time slots are 6AM–10AM, 10AM–2PM and 2PM–6PM. These slots require attention where more buses could be provided.

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Future Work Plans (for the upcoming week):

→ Make necessary corrections in the code for merging the CSV files (Excel & Python) & Crowd Risk Model.
→ To make sure that Crowd Risk Score is working properly and that the formula correctly considers the required values from other CSV/ Excel sheets. (Crowd Risk Score Categorization table)

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

2/5/2026
Signature of the student
(with date)

Signature of the Internal Guide
(with date)

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Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA

Register Number: 24MCA306

Date: 9/5/2026

Internal Guide's Name: MS. SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Make necessary corrections in the code for merging the CSV files (Excel & Python) and Crowd Risk Model Crowd risk score

→ To make sure that ~~time slot~~ is working properly and that the formula correctly considers the required values from other CSV/Excel sheets. (Crowd Risk Score categorization table)

Progress/Achievements for the current week:

→ Changes were made in the If-else loop to estimate the crowd risk level and classify them as Low, Medium, High.

→ A table of crowd risk score representation was created for classification to check the results.

0.0 – 2.4	Low	Minimal passenger congestion
2.5 – 3.5	Medium	Moderate passenger congestion
3.6 – 5.5	High	Heavy passenger congestion.

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Future Work Plans (for the upcoming week):

→ Create the dashboard in Tableau once after every observations/ranges are completely represented/shown in the Excel file.
→ observations here are the table of figures/distribution tables that help in understanding the data much better.

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

Lea Dsouza
09/05/2026
Signature of the student
(with date)

Signature of the Internal Guide
(with date)

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Yenepoya (Deemed to be University)

WEEKLY RESEARCH PROGRESS REPORT

Student Name: LEANDER CHRIS DSOUZA

Register Number: 24MCA306

Date: 16/05/2026

Internal Guide's Name: MS. SINDHU SANDESH

MAJOR PROJECT Title:

CROWD ESTIMATION AND TIME-SLOT RECOMMENDATION FOR MANGALORE KSRTC BUS SERVICES

Targets set for the current week:

→ Perform visualisation of Crowd Risk by Time slot, Route Level Risk & Crowd Risk Distribution using Tableau.
→ Prepared structured outputs for creating the dashboard.
→ Analyse route level patterns to determine safe travel periods.
→ Begin preparing for final outputs, report etc ——.

Progress/Achievements for the current week:

→ Successfully implemented the dashboard in Tableau with filtering and color palette.
→ Relevant bar graphs were used for representing Crowd Risk distribution by Time slot, Route Level Risk & Crowd Risk Distribution overall
→ Re-evaluated the working of the project so as to not provide misleading results.

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Future Work Plans (for the upcoming week):

→ Final report preparation ~~with~~ the final output/results ready for presentation, plagiarism checking etc ___

→ Observe the pattern of transport movement based on route, time ~~ones~~ at KSRTC bus stand, Bejai so that it matches the results of the dashboard.

Implementation shown: ☐ Yes ☐ No

Remarks by the Internal Guide:

Lea Dsouza
16/05/2026
Signature of the student
(with date)

Signature of the Internal Guide
(with date)

6. CONCLUSION OF THE RESEARCH AND FUTURE WORK

6.1 CONCLUSION

This research presented a timetable-driven and population-aware framework to analyse crowd intensity patterns in KSRTC bus services. Timetable data was integrated with district-level demographic indicators to estimate crowd intensity across different routes and time slots.

The analysis identified peak time slots and classified routes into Very Low, Low, Moderate, Medium, High and Very High crowd risk categories using rule-based scoring framework. Tableau visualizations supported the interpretation of route-level and district-level patterns.

The study shows that publicly available transport and demographic data can be used to support crowd intensity analysis without real-time passenger counting systems.

6.2 FUTURE WORK

Future work can improve the system by making use of:

1. Real-time passenger count data.
2. GPS-based vehicle tracking information.
3. Ticket booking and occupancy records.
4. Seasonal and festival travel patterns.
5. ML models for crowd prediction.
6. Integration with transport planning systems for service optimization.

These may improve the accuracy and usefulness of crowd intensity estimation in public transport systems.

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